

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech. (VLSI Design & Embedded Systems) (CBCS) Degree Examinations

May - June 2024

22VDE213 – SYNTHESIS AND OPTIMIZATION OF DIGITAL CIRCUITS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions each from **Unit - I and Unit - II** and **One full** question from **Unit - III**.

Unit – I

Marks BT* CO* PO*

1. a) Using Dijkstra's algorithm, find the shortest path in the graph shown in Fig.1(a). Show all steps.

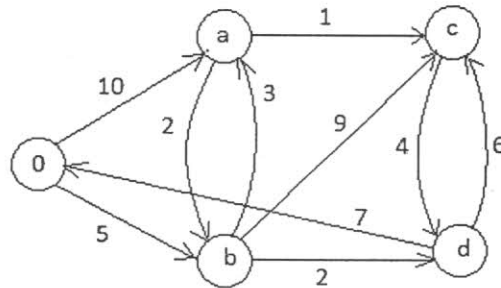


Fig.1(a)

- b) Write structural Verilog code for 4-bit parallel adder. Use full adder as component and also write Verilog code for it.
2. a) Let $f = a'b + b'c + ac'$. Evaluate the Boolean difference, consensus and smoothing of f with respect variable a .
- b) Consider the code segment

$xl = x + dx; ul = u - (3 * x * u * dx) - (3 * y * dx);$
 $yl = y + u * dx; c = xl < a;$

Calculate the lower bound on the total number of resources required to implement the code segment. Assume that all operations can be implemented by a general purpose ALU with unit execution delay.

3. a) Construct ROBDD for the functions (i) $f = (ab + bc) \text{ XOR } a$ using Shannon cofactoring (ii) $f = ac + bc$ using ITE operator. Use the variable order a, b, c .
- b) Consider the code segment

$xl = x + dx; ul = u - (3 * x * u * dx) - (3 * y * dx);$
 $yl = y + u * dx; c = xl < a;$

Schedule the operations using ASAP and ALAP scheduling algorithms. Also, calculate the mobility of operations and identify the operations in the critical path.

Unit – II

4. a) Compute the cover of complement of $f = wx + w + xz + xy + xyz$ using unate recursive paradigm.
- b) For the logic circuit shown in Fig.4(b), identify the false path and the true critical path and justify your answer. Assume that the propagation delay of each gate is 1 unit and inputs are available at time 0.

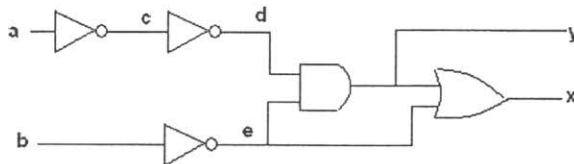


Fig.4(b)

5. a) Using Quine-McCluskey algorithm, find the minimum cover of $f(a, b, c, d) = \sum(0, 2, 4, 5, 6, 7, 8, 9, 10, 11, 13)$

P.T.O

- b) Consider the logic network shown in Fig.5(b) in which all NAND gates have propagation delay of 2 units and that of each inverter is 1 unit. The data ready times for all inputs is zero except for d where $t_d = 3$. Find the critical path delay. Decompose the network to reduce the delay. Also draw the resynthesized logic network.

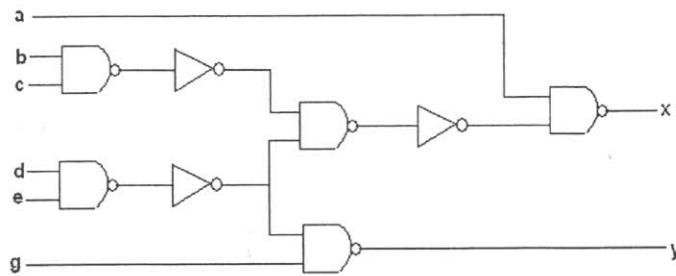


Fig.5(b)

10 L3 4 3

6. a) Consider the logic network defined by the following expressions:
 $f_x = ace + bce + de + g$; $f_y = ad + bd + cde + ge$; $f_z = abc$.
 Compute the kernels of f_x , f_y and f_z . Extract the single cube and multiple cube common sub expressions and optimize the network.
- b) A logic network is defined by the following equations:
 $p = eg$; $q = hk$; $r = c + d$; $s = p + q$; $t = lm$; $u = rs$;
 $w = bu$; $x = ar$; $y = b \oplus r$; $z = s + t$
 where $a, b, c, d, e, g, h, k, l, m$ are primary inputs and w, x, y, z are primary outputs. Draw the logic network. Find the topological critical paths assuming that all gates have unit delay and all input data-ready times are 0. Apply elimination of vertices to reduce delay.

10 L3 3 3

10 L3 4 3

Unit – III

7. a) Explain the transformations used in rule-based library binding with examples.
- b) Using tree covering algorithm find the area optimum technology mapping of the subject graph shown in Fig.Q7(b). If the following cells are available in a library: INV(1), NAND2(2), NOR2(3), OR2(4), AND2(3), AOI21(5), AOI22(6). (i) Draw the pattern graph for each of them. All cells are realized with NAND2 and INV gates, the number in the parenthesis is the cost of the cell. (ii) Show how the graph is covered optimally.

10 L2 5 3

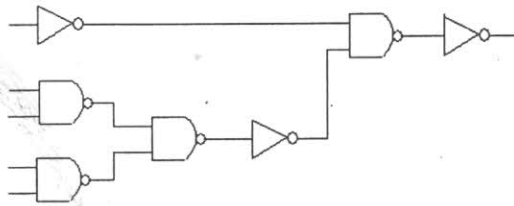


Fig.Q7(b)

10 L3 5 3

8. a) Consider the state table given below. Draw the state diagram, minimize the state table and draw the minimum state diagram. Also list the compatible states.

Present State	Next State		Output	
	x = 0	x = 1	x = 0	x = 1
S1	S3	S5	1	*
S2	S3	S5	*	1
S3	S2	S1	0	1
S4	S4	S5	0	1
S5	S4	S1	1	0

10 L3 5 3

- b) Explain different types of finite-state machine decomposition with diagrams.

10 L2 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
